MADHUMITHA S

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OBJECTIVE

To pursue an challenging position at Mphasis where I can apply my programming and problem-solving skills, contribute to software development projects, and continuously enhance my technical knowledge while growing with the organization..

EDUCATION

B.E. Electronics and Communication Engineering 2023–Present
K.S.R College of Engineering, Tiruchengode
CGPA: 8.6 (Up to 6th Semester)

Higher Secondary School 2022–2023
TAMM Govt Hr Sec School, Tiruppur
Percentage: 82%

Secondary School 2020–2021
Komarappa Sengunthar Girls Hr Sec School, Erode
Percentage: 90%

PROJECTS

Traffic Light Controller using Finite State Machine (FSM)
Tools Used: Verilog HDL, ModelSim

- Designed and implemented an FSM-based Traffic Light Controller using Verilog HDL.
- Developed state transition logic for Red, Yellow and Green signal sequencing.
- Simulated multiple traffic scenarios and verified functionality using ModelSim.
- Performed waveform analysis to validate timing and output behavior.

Design and Verification of 4-Bit Arithmetic Logic Unit (ALU)
Tools Used: Verilog HDL, ModelSim

- Designed and implemented a 4-bit ALU using Verilog HDL.
- Implemented arithmetic and logical operations including Add, Subtract, AND, OR and XOR.
- Developed testbenches for functional verification.
- Simulated and analyzed output waveforms using ModelSim.

INTERNSHIPS

Embedded IoT Intern – Tessolve

- Developed firmware using Bare Metal Programming on STM32 and MSP430 microcontrollers.
- Worked with GPIO, Timer, UART, I2C and SPI peripheral interfacing.
- Performed hardware debugging, validation and testing of embedded applications.

BSNL, Salem – Fundamentals of Telecommunication

- Gained knowledge of telecommunication systems, signal transmission and communication protocols.
- Understood network architecture, switching systems and telecom infrastructure.
- Observed the operation and maintenance of telecommunication equipment and network services.

Embedded C Programming – Embuzz Technologies, Salem

- Learned basics of Embedded C and microcontroller programming.
- Worked with AVR GPIO concepts and hardware interfacing fundamentals.
- Gained hands-on experience in debugging, testing and troubleshooting embedded applications.

CERTIFICATIONS

- Cloud Computing – NPTEL (SWAYAM), 2026 (65%) – Elite
- Advanced PCB Tools for Innovation – Chip Crafts Pvt Ltd
- Embedded Systems and IoT – Taras Systems and Solution
- Learnathon – ICT Academy

TECHNICAL SKILLS

- Programming: C, Java, Python Basics
- Web Technologies: HTML, CSS, JavaScript Basics
- Database: SQL, MySQL Basics
- Core Concepts: OOP, Data Structures Basics